Lecture 2: Number Systems & Arithmetic (MCQs)

1. UTF-32 encodes every character on ________ bytes.
a) 1 b) 2 c) 3 d) 4

• **Answer:** d) 4

2. Which of the following is a disadvantage of UTF-32?
a) It is variable length b) It is super wasteful in space c) It cannot represent English d) It is slower for text operations

• **Answer:** b) It is super wasteful in space

3. In _________, the Most Significant Byte (MSB) is stored in the lowest address location.
a) Little Endian b) Big Endian c) Middle Endian d) Dual Endian

• **Answer:** b) Big Endian

4. Intel x86 processors typically use _________ format.
a) Big Endian b) Little Endian c) Bi-Endian d) Neutral Endian

• **Answer:** b) Little Endian

5. If the 64-bit value 0x1122334455667788 is stored at address 9656 in a Little-Endian computer, the address of the byte 22 is _________.
a) 9656 b) 9657 c) 9662 d) 9663

• **Answer:** c) 9662 (Logic: 88->9656, 77->9657... 22->9662)

6. If the value 0xA1B2C3D4 is stored at address 0x20 in Big Endian, the value at address 0x20 is _________.
a) D4 b) C3 c) B2 d) A1

• **Answer:** d) A1

7. The decimal equivalent of the binary number (101100) is _________.
a) 44 b) 50 c) 34 d) 12

• **Answer:** a) 44

8. The binary equivalent of decimal (50) is _________.
a) 110010 b) 110011 c) 101010 d) 111010

• **Answer:** a) 110010

9. The decimal equivalent of Hex (1A5F) is _________.
a) 564 b) 6751 c) 4096 d) 32768

• **Answer:** b) 6751

10. To convert Hex A5 to binary, the result is ________.
a) 10100101 b) 10101010 c) 01011010 d) 11000101

• **Answer:** a) 10100101

11. In Sign-Magnitude representation, the MSB indicates ________.
a) The magnitude b) The sign (0 for pos, 1 for neg) c) The overflow d) The parity

• **Answer:** b) The sign (0 for pos, 1 for neg)

12. A problem with both Sign-Magnitude and 1's Complement is that ________.
a) They cannot represent negative numbers b) They have two representations for Zero c) They take more bits d) They are not supported by hardware

• **Answer:** b) They have two representations for Zero,

13. To find the 2's Complement of a number, you must ________.
a) Invert bits only b) Add 1 only c) Invert bits and add 1 d) Add 128

• **Answer:** c) Invert bits and add 1

14. The 2's complement of binary 11100011 is ________.
a) 00011100 b) 00011101 c) 11100010 d) 00011111

• **Answer:** b) 00011101

15. In 8-bit 2's complement, the value -5 is represented as ________.
a) 10000101 b) 11111010 c) 11111011 d) 00000101

• **Answer:** c) 11111011 (Invert 00000101 -> 11111010 + 1)

16. When extending a signed negative number (like 101) to 8 bits, the result is ________. a) 00000101 b) 10000101 c) 11111101 d) 01111101

• **Answer:** c) 11111101 (Sign Extension fills with 1s)

Converting binary fractions to decimal involves multiplying by ________. a) 2^n b) 2^{-n} c) .17
16^n (10^n d

12 2^{-n} (**Answer:** b •

18. The binary fraction 0.45 is converted to binary by ________.
a) Repeated division by 2 b) Repeated multiplication by 2 c) Adding 2 d) Subtracting 2

• **Answer:** b) Repeated multiplication by 2

19. Fixed-Point numbers are defined by two parameters: Width (w) and ________.
a) Sign bit (s) b) Magnitude (m) c) Binary point position (b) d) Scale factor (f)

• **Answer:** c) Binary point position (b)

20. In addition to 2's complement, 1 + 1 results in ________.
a) 0 with no carry b) 1 with no carry c) 0 with carry 1 d) 1 with carry 1

• **Answer:** c) 0 with carry 1

21. Overflow occurs when adding two numbers if ________.
a) You add a positive to a negative b) You add two positives and get a negative c) The result is zero d) There is a carry out

• **Answer:** b) You add two positives and get a negative

22. Can overflow occur when adding a positive number to a negative number?
a) Yes, always b) Yes, sometimes c) No, never d) Only in 1's complement

• **Answer:** c) No, never

23. The 'N' bit in the Condition Code Register is set if ________.
a) Result is Zero b) MSB of the result is 1 c) An overflow occurred d) A carry occurred

• **Answer:** b) MSB of the result is 1

24. The 'V' bit in the Condition Code Register stands for ________.
a) Valid b) Value c) Overflow d) Vector

• **Answer:** c) Overflow

25. If you perform 0110 + 1100 (where 1100 is -4), the result is 0010. The flags (N, Z, V, C) are: ________.
a) 0, 0, 0, 0 b) 1, 0, 0, 1 c) 0, 0, 0, 0, 1 d) 0, 1, 1

• **Answer:** c) 0, 0, 0, 1 (Result is +2, Carry out is 1, No overflow)

26. The 1's complement of 10001 is _______.
a) 01111 b) 01110 c) 11110 d) 00001

• **Answer:** b) 01110

27. To represent 6.75 in Fixed-Point (4 integer bits, 4 fraction bits), the binary is ________.
a) 0110.1100 b) 0110.1010 c) 0110.0110 d) 0111.1100

• **Answer:** a) 0110.1100 (,)

28. Which of the following numbers would cause an Overflow if added to 0111 (+7) in a 4-bit signed system?
a) 1111 (-1) b) 0001 (+1) c) 1000 (-8) d) 1100 (-4)

• **Answer:** b) 0001 (+1) (Result 1000 is -8, which is invalid for +7 + +1)

29. The decimal equivalent of (233)10 in binary is ________.
a) 00100101 b) 11101001 c) 00101100 d) 10101010

• **Answer:** b) 11101001 (From Exam 2025)

30. In signed integers, if the highest bit is 1, then it indicates that the number is ________. a) Positive b) Negative c) does not matter d) invalid

• **Answer:** b) Negative (From Exam 2025)